VHDL Assignment 3

Gate-level circuits, synthesized designs and simulation evidence

Five sequential circuits | GHDL 5.0.1 | Logisim 2.7.1 | VHDL-2008

Verification method

Each circuit has two independently executed implementations: a VHDL design compiled, elaborated, simulated and synthesized with GHDL, and an individual-gate Logisim project loaded into the actual Logisim simulation engine. No built-in latch or flip-flop components are used in the Logisim designs. Named tunnels connect signals and feedback paths electrically.

Results

Circuit	VHDL simulation	Synthesis	Logisim steps
RS Latch	PASS	PASS	6
D Latch with Enable	PASS	PASS	6
Positive-edge D Flip-Flop	PASS	PASS	8
Positive-edge JK Flip-Flop	PASS	PASS	13
D Flip-Flop with Asynchronous Set/Clear	PASS	PASS	17

48 expected-Q checks pass in each implementation; two initial preparation steps have no specified power-on Q. Six additional Qn checks verify the RS latch. The tests cover forbidden RS inputs, latch transparency/hold, rising and falling edges, all JK modes, asynchronous controls with both clock levels, priority and release behavior.

Scope and timing assumptions

These are ideal digital functional simulations, not transistor-level timing proofs. Uninitialized storage has no guaranteed power-on value. Input values are held stable before sampling edges. Real hardware must satisfy setup/hold and asynchronous recovery/removal timing; asynchronous inputs must not be released near a clock edge. The RS forbidden-state release is deliberately not assigned a deterministic result. Feedback latches and dual asynchronous controls may need device-specific primitives for a particular FPGA target; synthesis here is generic GHDL synthesis, without FPGA place-and-route.

Files and reproduction

vhdl/: five sources; tb/: assertion-based testbenches; logisim/: five projects; results/: VCD files, compiled synthesis netlists, logs and vectors; schematics/: actual Logisim renders and synthesized-netlist diagrams; scripts/: repeatable verification. See README.md for commands. The generated synthesis diagrams visualize assignments and storage in the emitted netlist; they are not FPGA technology-mapped schematics.

1. RS Latch

Gate-level implementation

Cross-coupled NOR gates implement Q = NOT(R OR Qn), Qn = NOT(S OR Q). S,R = 10 sets; 01 resets; 00 holds. The forbidden 11 condition forces Q=Qn=0, so the outputs are not complementary. Simultaneous release of 11 to 00 has no defined physical result; the test recovers by applying a valid reset.

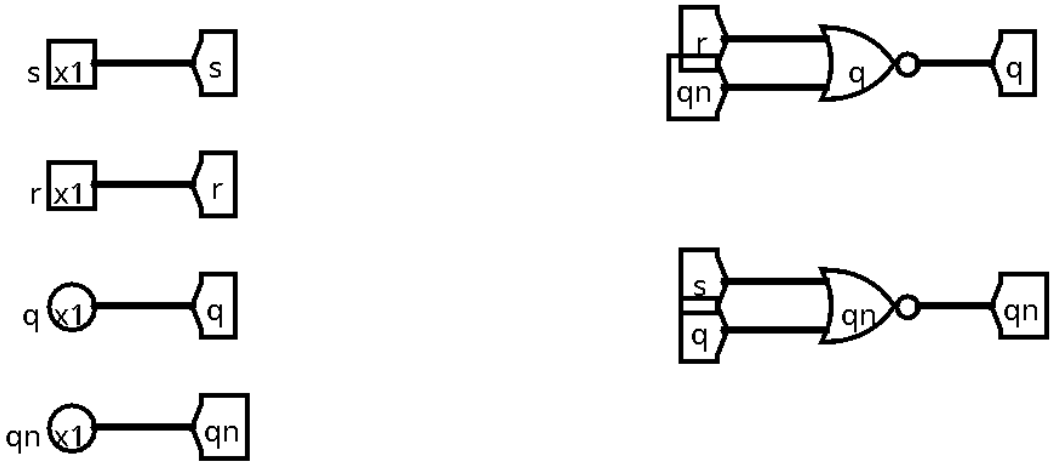


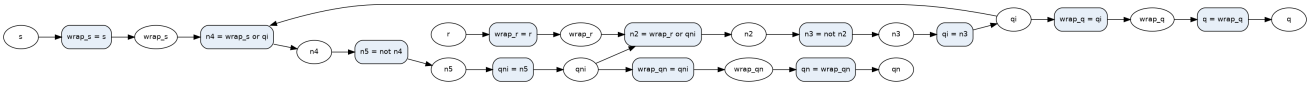
Figure: rendered directly from the submitted .circ project using Logisim. Matching tunnel labels denote connected nets; all storage feedback is explicit at gate level. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Reset	0	0	PASS
2	Hold reset	0	0	PASS
3	Set	1	1	PASS
4	Hold set	1	1	PASS
5	Forbidden both outputs zero	0	0	PASS
6	Recover by reset	0	0	PASS

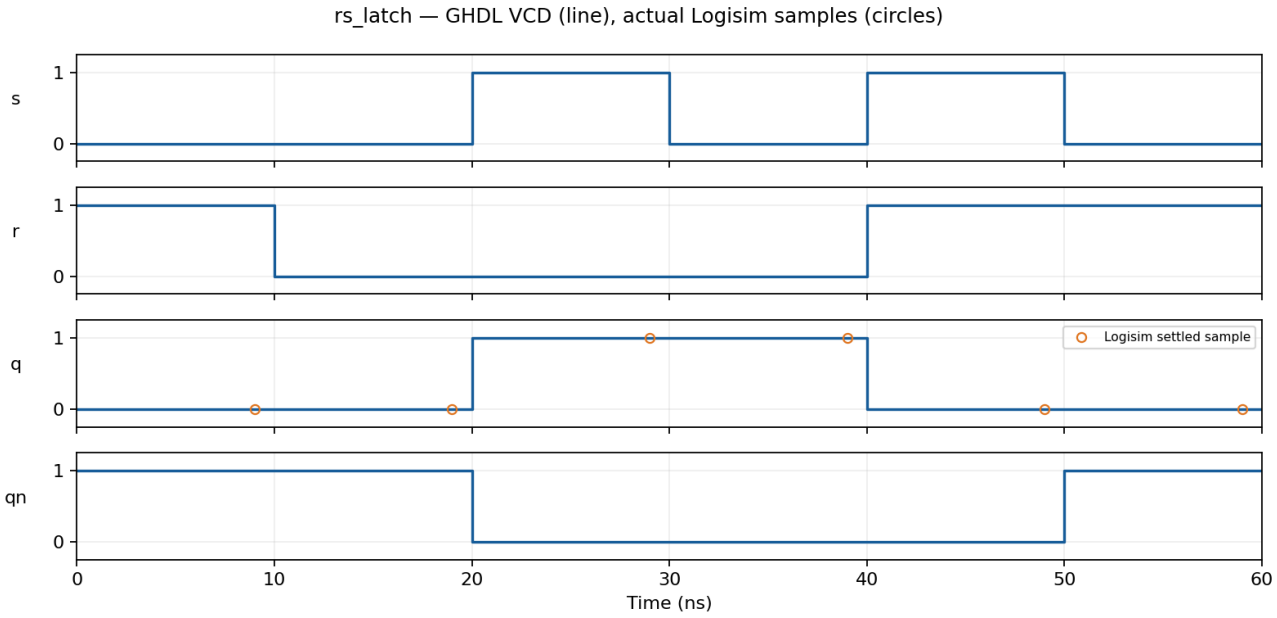
1. RS Latch — VHDL

Generated synthesis schematic



Generated from results/rs_latch_synth.vhd; boxes represent emitted concurrent assignments or sequential processes. See the text netlist for exact expressions.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_rs_latch.vhd:47:3:@60ns:(report note): PASS rs_latch
simulation stopped @60ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/rs_latch.vhd. Testbench: tb/tb_rs_latch.vhd.

2. D Latch with Enable

Gate-level implementation

A NAND inverter creates Dn. Sbar=NAND(D,en) and Rbar=NAND(Dn,en) drive a cross-coupled NAND latch. When en=1, Q follows D; when en=0, both latch inputs become 1 and the previous state is held.

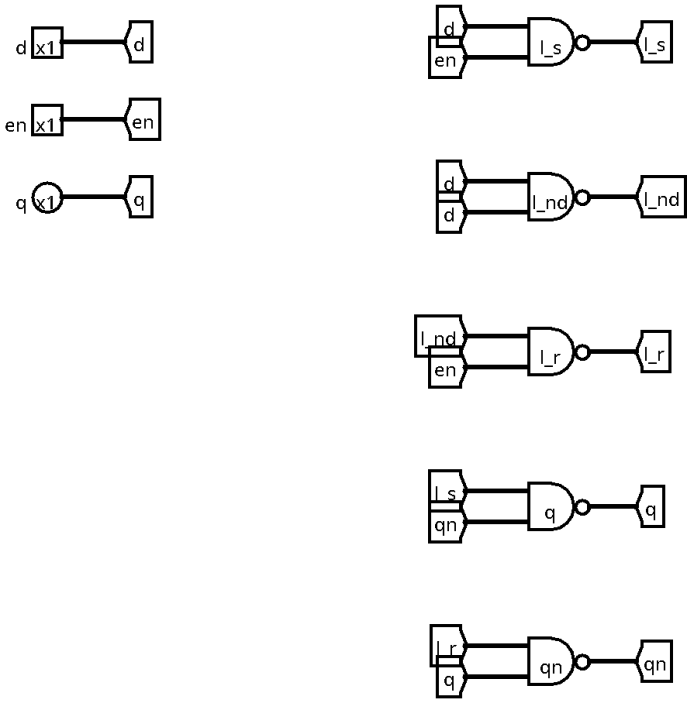


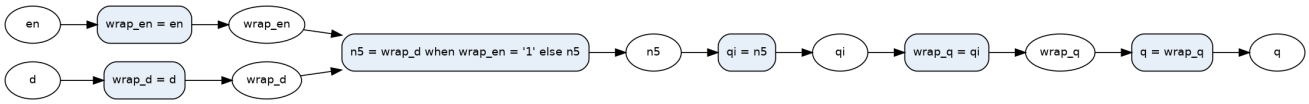
Figure: rendered directly from the submitted .circ project using Logisim. Matching tunnel labels denote connected nets; all storage feedback is explicit at gate level. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Initialize	0	0	PASS
2	Transparent high	1	1	PASS
3	Transparent low	0	0	PASS
4	Close	0	0	PASS
5	Hold despite D change	0	0	PASS
6	Reopen	1	1	PASS

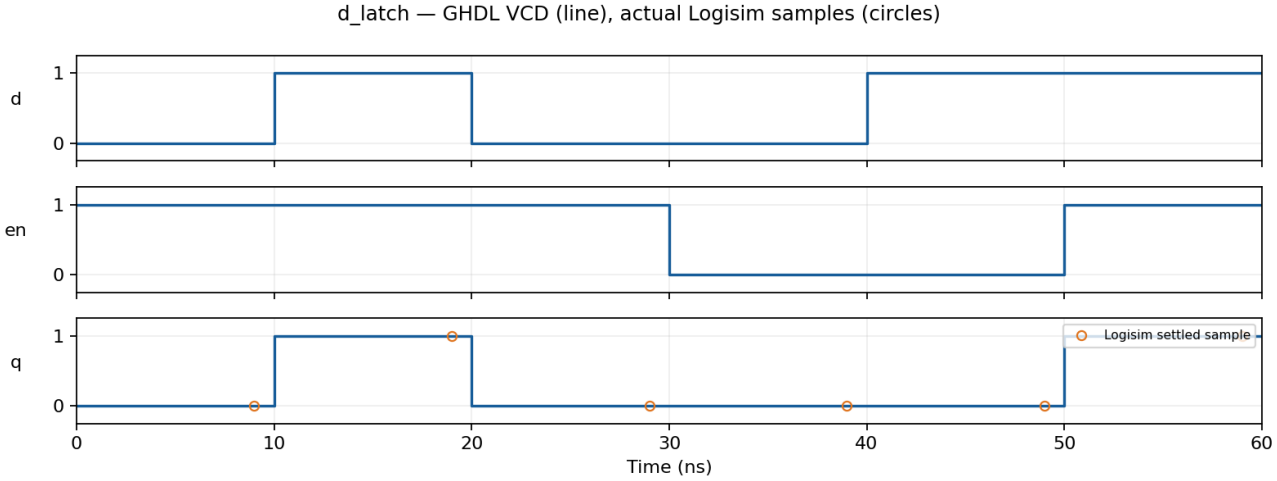
2. D Latch with Enable — VHDL

Generated synthesis schematic



Generated from results/d_latch_synth.vhd; boxes represent emitted concurrent assignments or sequential processes. See the text netlist for exact expressions.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_d_latch.vhd:38:3:@60ns:(report note): PASS d_latch
simulation stopped @60ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_latch.vhd. Testbench: tb/tb_d_latch.vhd.

3. Positive-edge D Flip-Flop

Gate-level implementation

Two gate-built D latches form a master-slave flip-flop. The master is enabled by NOT(clk), and the slave by clk. The master samples during the low phase; the slave transfers that stored value on the rising edge. Changes to D during the high phase do not reach Q.

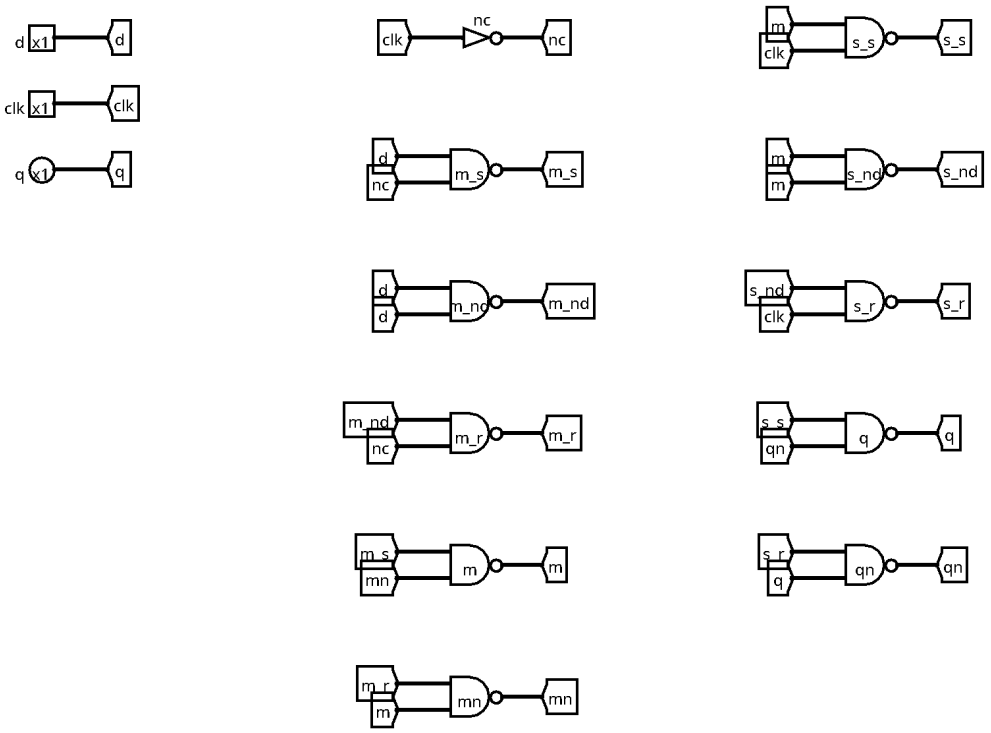


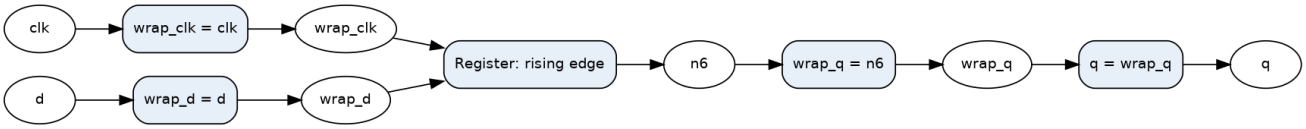
Figure: rendered directly from the submitted .circ project using Logisim. Matching tunnel labels denote connected nets; all storage feedback is explicit at gate level. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Initialize low phase	E	-	PASS
2	Rising capture zero	0	0	PASS
3	High phase hold	0	0	PASS
4	Falling edge hold	0	0	PASS
5	Rising capture one	1	1	PASS
6	High phase hold one	1	1	PASS
7	Falling hold one	1	1	PASS
8	Rising capture zero	0	0	PASS

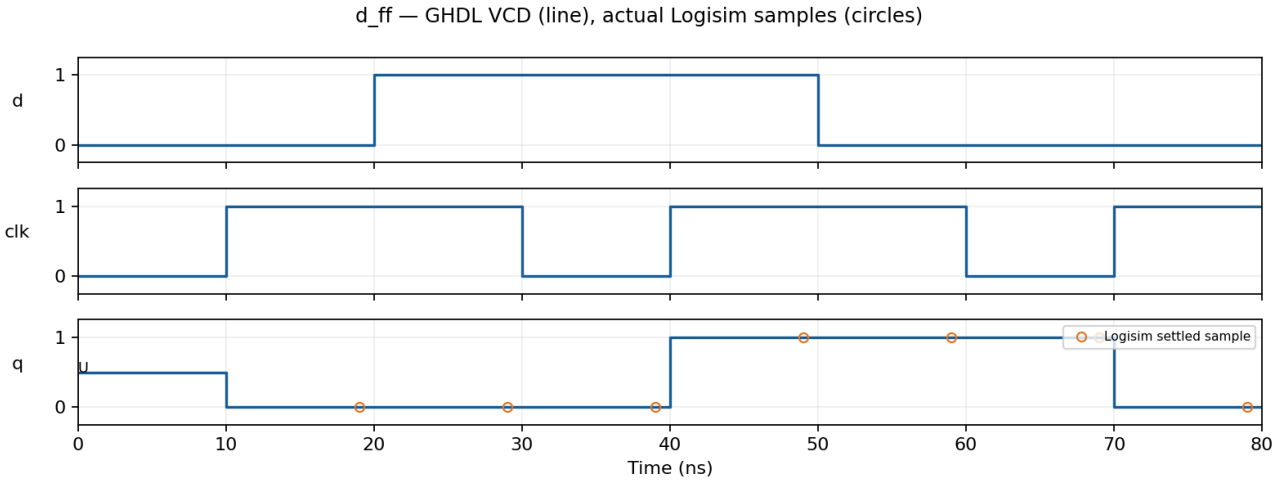
3. Positive-edge D Flip-Flop — VHDL

Generated synthesis schematic



Generated from results/d_ff_synth.vhd; boxes represent emitted concurrent assignments or sequential processes. See the text netlist for exact expressions.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_d_ff.vhd:43:3:@80ns:(report note): PASS d_ff
simulation stopped @80ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_ff.vhd. Testbench: tb/tb_d_ff.vhd.

4. Positive-edge JK Flip-Flop

Gate-level implementation

The feedback next-state equation is Dnext=(J AND Qn) OR ((NOT K) AND Q). NAND gates implement this equation and two NAND D latches form the master-slave storage. JK=00 holds, 01 resets, 10 sets, and 11 toggles once per rising edge.

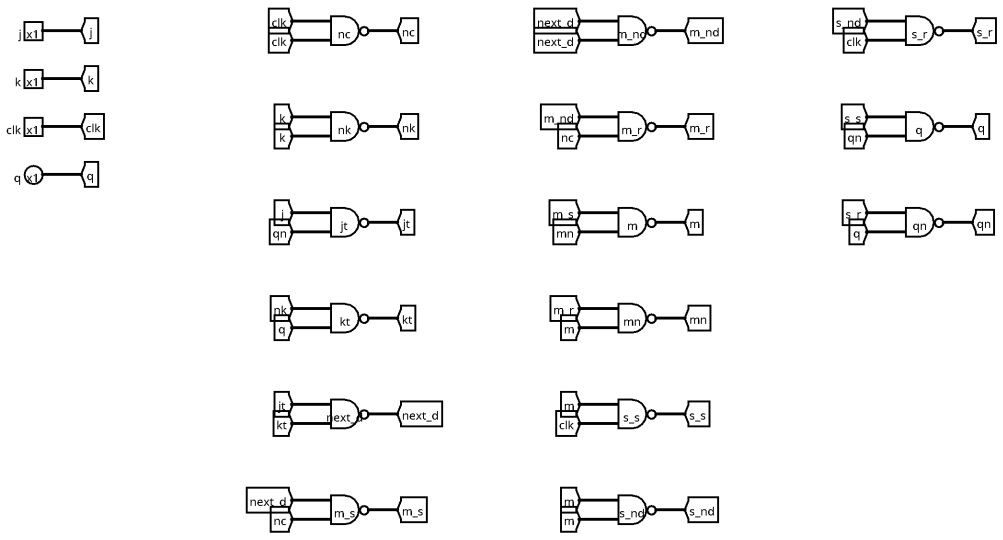


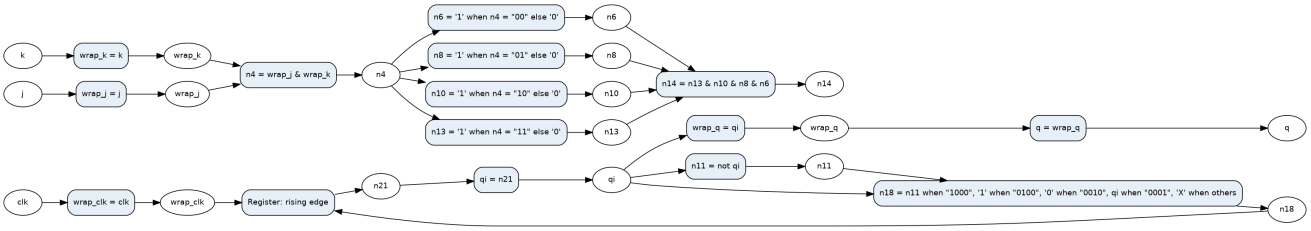
Figure: rendered directly from the submitted .circ project using Logisim. Matching tunnel labels denote connected nets; all storage feedback is explicit at gate level. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Prepare reset	E	-	PASS
2	Reset	0	0	PASS
3	Prepare hold	0	0	PASS
4	Hold zero	0	0	PASS
5	Prepare set	0	0	PASS
6	Set	1	1	PASS
7	Prepare hold one	1	1	PASS
8	Hold one	1	1	PASS
9	Prepare toggle	1	1	PASS
10	Toggle to zero	0	0	PASS
11	Falling hold	0	0	PASS
12	Toggle to one	1	1	PASS
13	High phase hold	1	1	PASS

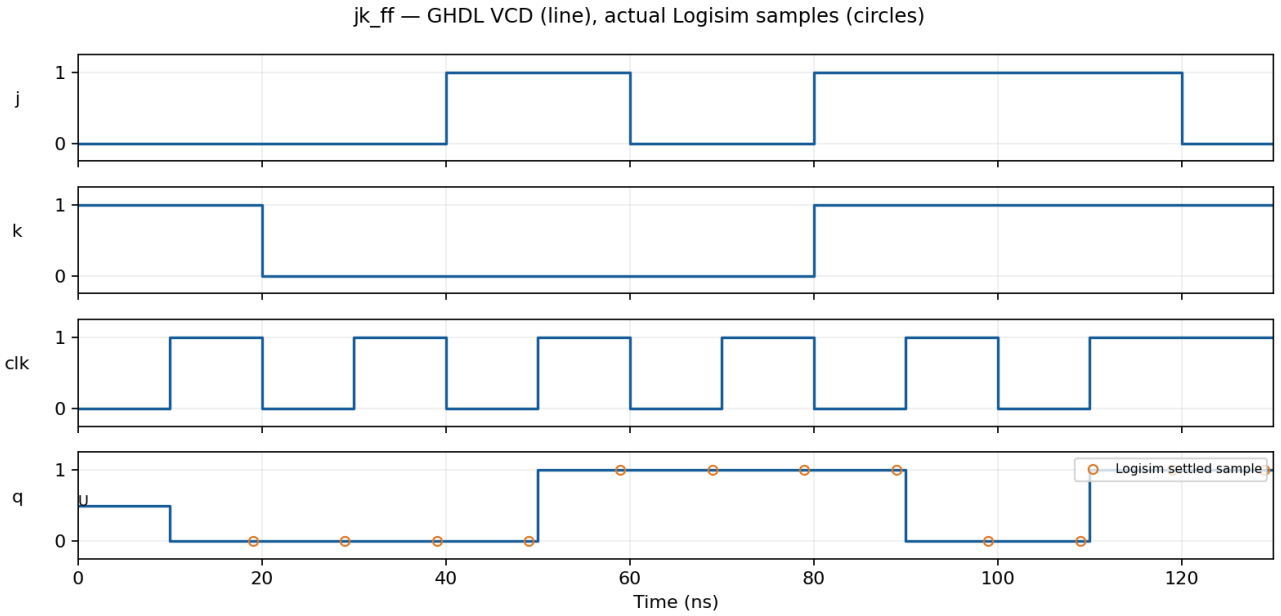
4. Positive-edge JK Flip-Flop — VHDL

Generated synthesis schematic



Generated from results/jk_ff_synth.vhd; boxes represent emitted concurrent assignments or sequential processes. See the text netlist for exact expressions.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_jk_ff.vhd:73:3:@130ns:(report note): PASS jk_ff
simulation stopped @130ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/jk_ff.vhd. Testbench: tb/tb_jk_ff.vhd.

5. D Flip-Flop with Asynchronous Set/Clear

Gate-level implementation

Both master and slave use gate-built NOR storage with asynchronous overrides. run=NOT(set OR clr), effective_set=set AND NOT(clr). For each latch, S=(D AND enable AND run) OR effective_set, R=(NOT(D) AND enable AND run) OR clr. Thus clr always wins. Applying controls to both stages prevents a stale master value from appearing when the controls are released during a high clock phase.

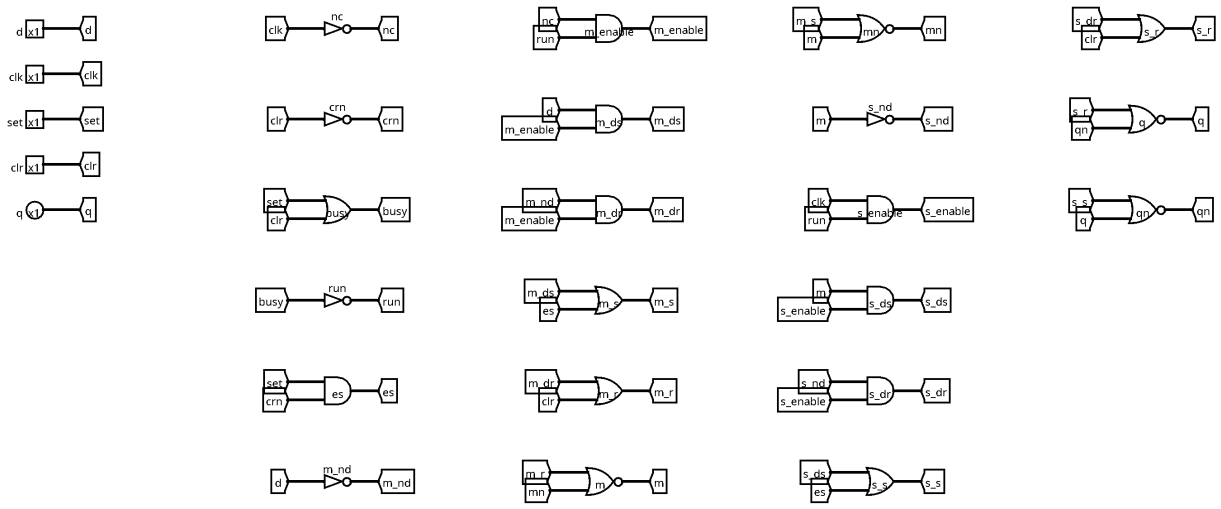


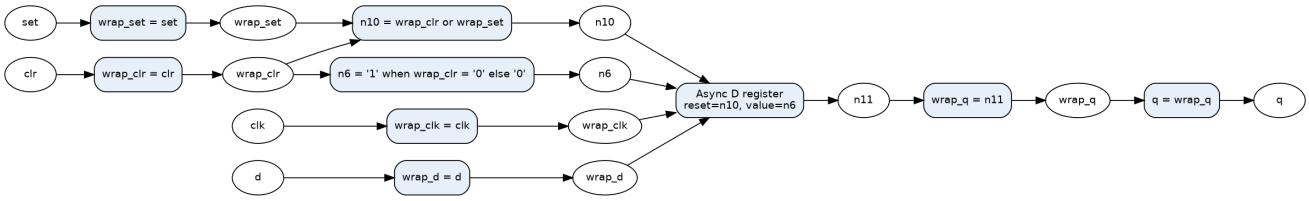
Figure: rendered directly from the submitted .circ project using Logisim. Matching tunnel labels denote connected nets; all storage feedback is explicit at gate level. The displayed states are hidden to emphasize connectivity.

Simulation checks

Step	Input case	Q	Expected	Result
1	Async clear low clock	0	0	PASS
2	Release clear	0	0	PASS
3	Async set low clock	1	1	PASS
4	Clear priority both active	0	0	PASS
5	Set remains asserted	1	1	PASS
6	Release set	1	1	PASS
7	Normal capture zero	0	0	PASS
8	High phase data hold	0	0	PASS
9	Async set high clock	1	1	PASS
10	Clear priority high clock	0	0	PASS
11	Release controls high clock	0	0	PASS
12	Falling hold	0	0	PASS
13	Normal capture one	1	1	PASS
14	Async clear high clock	0	0	PASS
15	Release clear high clock	0	0	PASS
16	Prepare zero	0	0	PASS
17	Capture zero again	0	0	PASS

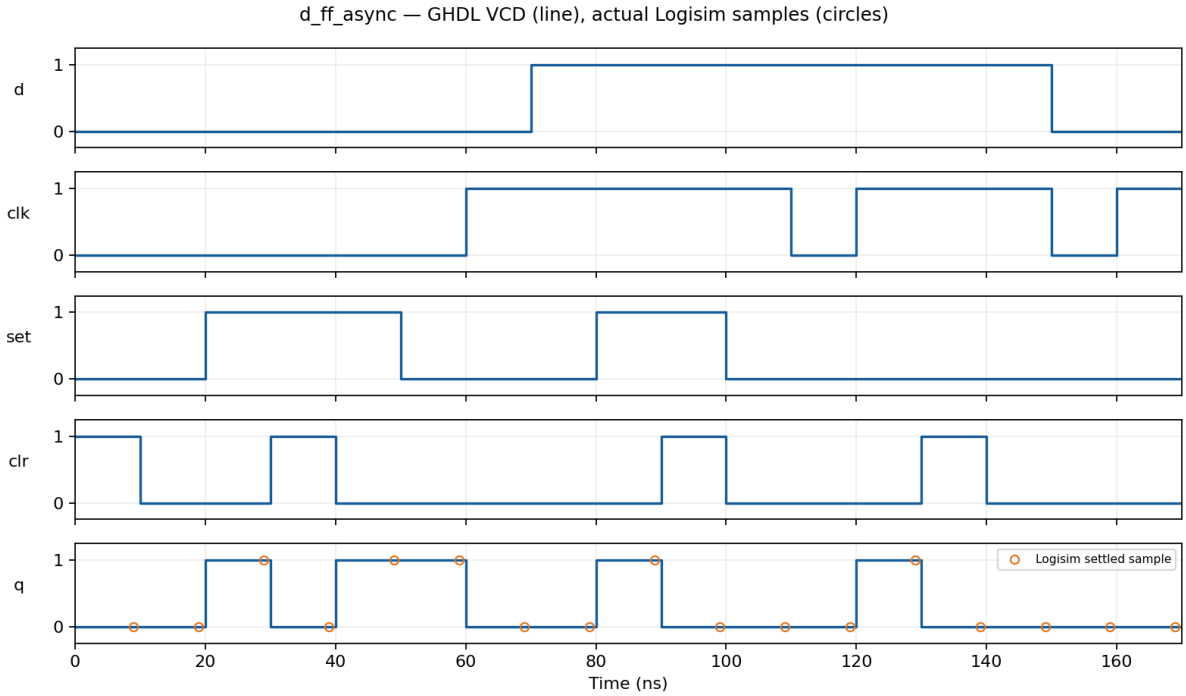
5. D Flip-Flop with Asynchronous Set/Clear — VHDL

Generated synthesis schematic



Generated from results/d_ff_async_synth.vhd; boxes represent emitted concurrent assignments or sequential processes. See the text netlist for exact expressions.

Simulation waveform



Blue: GHDL VCD traces. Orange circles: independently simulated Logisim Q after each input step (placed 1 ns before the corresponding VHDL step boundary for readability). Unknown initial values are marked U/X or shown at midlevel. Logisim time is propagation settling, so the circles compare logical states, not propagation delay.

VHDL verification

/workspace/output/assignment3/tb/tb_d_ff_async.vhd:84:3:@170ns:(report note): PASS d_ff_async simulation stopped @170ns

All RTL and synthesized-netlist assertions passed; generic synthesis completed successfully. Source: vhdl/d_ff_async.vhd. Testbench: tb/tb_d_ff_async.vhd.